Minh Le

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Education

Drexel University Philadelphia, PA

B.S in Software Engineering, Minor in Finance | GPA: 3.79/4.0 *Expected May 2028*

- Coursework: Object-Oriented Programming (Java), Data Structures & Algorithms, Software Architectures, Database Systems, Artificial Intelligence, Statistics & Probability
- Affiliations: Hack4Impact (Developer), ASME (Technical Coordinator), Drexel AI, Stanford's CiP

Technical Skills

Languages: Go (Golang), Python, C/C++, Java, JavaScript/TypeScript, YAML, SQL, Bash, HTML/CSS
Frameworks: Spring Boot, Next.js, Angular, JUnit, Hono
Libraries & Tools: Claude Code, Git, Jira, React, Node.js, Pandas, Cloudflare, Kafka, Azure DevOps
Databases & Infrastructure: PostgreSQL, MySQL, MongoDB, Docker, Kubernetes, Linux, AWS, Azure, Helm

Experience

VisioneerIT Mar 2026 – Sept 2026

Software Engineer Intern *Baltimore, MD*

- Built **Next.js/React** components and tier-gated APIs across multiple government data sources, giving users filtered access to **100k+** government agencies and bids.
- Engineered a 5-stage **RAG** hybrid search pipeline using BGE, BM25 and CE reranking across **87,000+** indexed government policy documents, achieving sub-600ms query latency and **85.9%** average relevance score.
- Developed automated ETL pipeline to scrape active government solicitations across **all 50** states, configuring schema mappings and deduplication logic for **MySQL** tables.

Bentley Systems Mar 2025 – Sept 2025

Software Engineer Intern *Philadelphia, PA*

- Modernized legacy **RPC** protocols to **REST** endpoints with session affinity using cookies, reducing response times from ∼**2s** to **0.2s** for returning users across a Go-based Kubernetes backend.
- Deployed an automated **vertical scaling** pipeline across 8+ regions using an open-source recommendation engine, generating right-sizing reports that improved utilization by **35%** and saved ∼**$2,000** annually per node.
- Engineered **Grafana** dashboards and alert rules for **20+ microservices**, enabling proactive failure detection to protect **service level agreements** (SLAs) and prevent costly downtime.

Drexel University Jun 2024 – Sept 2024

Research Assistant *Philadelphia, PA*

- Developed a specialized full-stack **NextJS** web platform for analyzing stigma in drug-related content, integrating **Hono**, **Docker**, and **MongoDB** to accelerate data processing workflows for researchers.
- Built and deployed custom LLM workflows integrating **OpenAI GPT-4** for advanced NLP content analysis, preserving users' stylistic profiles with over **90%** classification accuracy.

Projects & Open Source

Pkmn-Quant - Algorithmic Trading Backtester | *Python, Polars, C++, scikit-learn, DuckDB*

- Built an event-driven backtester over a **3M-row** Parquet price warehouse with **T+1** fills, **12.75%** fees, and tiered liquidity caps, enabling all **5** strategies to generate live daily trade signals with **zero** code changes.
- Trained a gradient-boosted model to rank assets by predicted forward returns with **no look-ahead bias**, posting a **+6.0%** out-of-sample return that beat **3 of 4** hand-built strategies under **11-fold** walk-forward analysis.
- Profiled and rewrote the engine's per-day pricing hot path, cutting backtest runtime **1.9x** (3.8s to 2.0s) with zero behavior drift, verified byte-identical by golden regression tests in a **267-test** CI suite.

OpenTelemetry (Open Source Contributor) | *Go, C, eBPF, OTLP, Docker*

- Introduced a new **OTLP Logs** export pipeline for an **eBPF**-based **Go/C** auto-instrumentation agent, extracting kernel-level request timings to correlate natively with parent traces via shared **span IDs**.
- Implemented an opt-in configuration toggle that consolidates **3** verbose sibling spans into **1** unified log record, reducing trace export volume by **66%** while maintaining **100%** backward compatibility.

Harbor (Open Source Contributor) | *Go, PostgreSQL, Keycloak, OIDC, Angular*

- Developed an OIDC group-based authentication filter for a CNCF-graduated Go backend, integrating Keycloak JWT parsing to secure system access across API, Docker CLI, and Angular frontend interfaces.
- Added a multi-layer configuration pipeline from a **PostgreSQL** registry, achieving 100% backward compatibility while validating authorization states across 8 edge cases to guarantee zero security regressions.

SightLink - CodeFest 2026 | *Python, MediaPipe, MobileNetV2, TensorFlow, OpenCV*

- Developed a real-time ASL fingerspelling translator using a confidence hybrid classifier, routing 30fps webcam frames through a fast landmark **MLP** (97.5% accuracy) with **CNN** fallback, achieving ∼99.9% accuracy.
- Trained both models on 87,000 Kaggle ASL images across 29 classes, implementing scale-invariant landmark normalization (wrist-centered, bounding-box-scaled) that boosted MLP accuracy 15% above projected baseline.

Dragon Shell - Custom Shell and TCP Server | *C, POSIX Sockets, pthreads, Linux*

- Built a UNIX shell in **C**, that runs piped commands and I/O redirection using fork, execvp, and dup2, cleaning up child processes with waitpid.
- Extended shell into a multi-threaded **TCP** server in C, spawning a thread per client connection, allowing multiple users to run remote commands at once.

SocialGym - ML Chat Platform | *React, Spring Boot, Python, Kafka, PostgreSQL*

- Implemented an event-driven ML pipeline via **Kafka**, allowing a Python microservice to perform asynchronous Hugging Face sentiment analysis, maintaining **100%** main-thread availability during heavy NLP workloads.
- Engineered a low-latency, full-duplex chat interface using **WebSockets** and React, dynamically updating the UI with real-time AI feedback without HTTP polling lag, ensuring **sub-100ms** message delivery.

Stylify - Virtual Wardrobe App | *React+Vite, Hono, PostgreSQL, AWS Lambda, S3, CloudFront*

- Built a responsive virtual wardrobe system using **React+Vite and PostgreSQL**, featuring complete CRUD functionality to manage **500+** garment entries and generate recommendations using multi-param filtering.
- Migrated deployment from EC2 to a serverless architecture using **AWS Lambda, ECR, CloudFront+S3**, cutting infrastructure costs from **$12/mo** to **$0.15/mo** while maintaining **99.9%** uptime.

DU ProfessorView - Rate My Professor Extension | *TypeScript, Chrome MV3, WXT, Cloudflare Workers*

- Built a chrome extension adding Rate My Professors ratings to Drexel's course catalog, used by **300** students.
- Designed a lookup that checks a local browser cache, then a **Cloudflare Workers** + KV shared cache.